

CS331 Error Detection: Practice Sheet

A. Simple Parity Check — 3 Questions

Assume **even parity** unless stated otherwise.

Q1 — Sender

Encode each 7-bit dataword into an 8-bit codeword by adding the parity bit:

- a) 1011010
- b) 1110001
- c) 0001011

For each one, show the number of 1s, parity bit, and final codeword.

Q2 — Receiver

The following codewords were received using even parity. Determine whether each is **valid or corrupted**:

- a) 10101010
- b) 11010111
- c) 00110011
- d) 11111110

Q3 — Odd parity

A system uses **odd parity**. Encode:

- a) 1011001
- b) 1111111
- c) 0000000

Then explain what the receiver expects the total number of 1s to be.

B. Two-Dimensional Parity — 3 Questions

Use **even parity**.

Q4 — Sender

Calculate the **row parity bits and column parity bits**:

1	0	1	1
0	1	0	1
1	1	1	0
0	0	1	1

Then construct the complete transmitted block.

Q5 — Receiver: Locate the Error

The following block uses even two-dimensional parity:

1	0	1	1		1
0	1	1	1		0
1	1	1	0		1
0	0	1	1		0

0	0	1	1		

One data bit has been corrupted.

Determine:

1. Which row fails?
2. Which column fails?
3. Which exact bit is corrupted?
4. What should that bit be?

Q6 — Multiple Errors

Suppose a two-dimensional parity block experiences:

- a) 1-bit error
- b) 2-bit errors
- c) 3-bit errors
- d) 4-bit errors

For each, state whether two-dimensional parity is guaranteed to detect it according to your slides.

This distinction matters because your slides specifically state that it can detect **one, two, and three errors, but not all four-error patterns**, while a single-bit error can be located and corrected.

C. Internet Checksum — 3 Questions

Your course uses **one's-complement addition**, including end-around carry.

Q7 — Sender

Calculate the **8-bit checksum**:

```
Word 1: 11001100
Word 2: 10101010
```

Show:

```
Addition
→ End-around carry (if any)
→ Sum
→ Complement
→ Checksum
→ Complete transmitted packet
```

Q8 — Sender

Calculate the checksum:

```
Word 1: 11110000
Word 2: 10110101
```

Again, show the carry handling and final transmitted packet.

Q9 — Receiver

The receiver gets:

```
Word 1: 10101001
Word 2: 00111001
Checksum: 00011101
```

Perform the receiver calculation and decide:

Error-free or corrupted?

Remember the receiver adds **data + checksum**. In your slides, a correct packet produces all 1s before complementing, and therefore all 0s after complementing.

D. CRC — Main Focus

This should get most of your practice time.

For **every sender question**, show:

1. Generator length
2. Number of zeros appended
3. Dividend
4. Modulo-2 division
5. Remainder / CRC
6. Final transmitted codeword

Our slides explicitly use **modulo-2 binary division** and call the CRC divisor the **generator polynomial/generator**.

Q10 — CRC Warm-Up

```
Dataword: 1001
Generator: 1011
```

Calculate the CRC and transmitted codeword.

Q11 — CRC Sender

Dataword: 1101
Generator: 101

Calculate the complete transmitted codeword.

Q12 — CRC Sender

Dataword: 101101
Generator: 1101

Calculate:

Appended zeros = ?
Dividend = ?
CRC = ?
Codeword = ?

Q13 — CRC Sender

Dataword: 110101
Generator: 1011

Perform the complete sender calculation.

Q14 — Generator Polynomial

The generator polynomial is:

$$X^3 + x + 1$$

Answer:

- a) What is its binary representation?
- b) What is its degree?
- c) How many CRC bits are produced?
- d) How many zeros are appended to the dataword?

Our slides actually use this exact polynomial-to-generator relationship.

Q15 — Generator Polynomial

Convert:

$$x^4 + x^2 + 1$$

into its binary generator.

Then answer:

Generator = ?
Degree = ?
Number of appended zeros = ?
Length of CRC remainder = ?

CRC Receiver Questions

Now switch perspectives.

At the receiver:

Divide the entire received codeword by the generator.

Q16 — Receiver

```
Received: 1001110  
Generator: 1011
```

Perform modulo-2 division.

Then state:

```
Remainder = ?  
Error detected? YES / NO
```

Q17 — Receiver

```
Received: 1101001  
Generator: 101
```

Perform the receiver-side division and determine whether the received data passes the CRC check.

Q18 — Receiver

```
Received: 101101001  
Generator: 1101
```

Calculate the remainder and decide whether the frame should be accepted.

CRC Sender → Receiver Combined

These are the ones I'd especially practice.

Q19 ☆ ☆

Sender has:

```
Dataword: 101011  
Generator: 1011
```

Sender

Find:

```
CRC = ?  
Transmitted codeword = ?
```

Receiver

Take your transmitted codeword and divide it by 1011.

```
Receiver remainder = ?
```

 Explain why you should expect that result.

Q20 ★★

Sender:

```
Dataword: 111001  
Generator: 1101
```

Calculate the transmitted codeword.

Then suppose **one bit changes during transmission**.

Choose one bit in your transmitted codeword and flip it:

```
0 → 1  
or  
1 → 0
```

 At the receiver, divide the corrupted codeword by 1101.

Determine:

```
Receiver remainder = ?  
Error detected? YES / NO
```

CRC Polynomial + Calculation Challenge

Q21 ★★★

Given:

```
Dataword: 1011011  
Generator polynomial:  $x^3 + x^2 + 1$ 
```

Do everything:

1. Convert polynomial $\rightarrow$ binary generator.
 2. Determine number of zeros to append.
 3. Append the zeros.
 4. Perform modulo-2 division.
 5. Find CRC.
 6. Construct transmitted codeword.
 7. Verify the codeword at the receiver.
-

Q22 ★★

Given:

```
Dataword: 1100101
Generator polynomial:  $x^4 + x + 1$ 
```

Find:

```
Binary generator
Number of appended zeros
Dividend
CRC
Transmitted codeword
```

Then verify the transmitted codeword at the receiver.

CRC Concept Questions

Q23

During CRC division, you need to calculate:

```
1011
XOR
1101
----
????
```

Calculate the result.

Q24

A generator contains **5 bits**.

Answer:

```
How many zeros are appended? _____
How many bits are in the CRC? _____
```

Q25

A sender calculates:

```
Dataword: 101101
CRC:      011
```

What is the transmitted codeword?

Q26

The receiver divides the complete received codeword by the generator and gets:

```
000
```

What does this mean?

Q27

The receiver instead gets:

Exam-Style Mixed Question**Q28**

For each situation, identify the error-detection method:

- A.** One redundant bit is added so the total number of 1s is even.
- B.** Data is arranged into rows and columns and parity is calculated in both directions.
- C.** A generator polynomial is converted into a binary divisor and modulo-2 division is performed.
- D.** Multiple words are added using one's-complement arithmetic and the result is complemented.

Choose from:

```
Simple Parity
Two-Dimensional Parity
CRC
Checksum
```

What I'd prioritize

Given what your doctor said, I wouldn't split your time evenly. I'd do the three **simple parity** questions, all three **2D parity** questions, all three **checksum** questions, and then spend the majority of the session on **Q10–Q22 for CRC**.

For CRC in particular, make sure you can do these four things without notes:

polynomial → binary generator → sender division → receiver verification.

That is the full CRC workflow our slides are building toward.

CS331 Error Detection — Answer Sheet

I checked the CRC calculations separately so the remainders/codewords below are consistent.

A. Simple Parity Check

Q1 — Even parity sender

Dataword	# of 1s	Parity bit	Codeword
1011010	4	0	10110100
1110001	4	0	11100010
0001011	3	1	00010111

Q2 — Receiver

Received	# of 1s	Result
10101010	4	✓ Valid
11010111	6	✓ Valid
00110011	4	✓ Valid
11111110	7	✗ Corrupted

Q3 — Odd parity

Dataword	# of 1s	Parity	Codeword
1011001	4	1	10110011
1111111	7	0	11111110
0000000	0	1	00000001

For odd parity, the receiver expects the **total number of 1s to be odd**.

B. Two-Dimensional Parity

Q4 — Sender

Original:

1	0	1	1
0	1	0	1
1	1	1	0
0	0	1	1
1			
0			

Row parity:

Column parity:

0	0	1	1
---	---	---	---

Complete block:

1	0	1	1		1
0	1	0	1		0
1	1	1	0		1
0	0	1	1		0

0	0	1	1		

Q5 — Locate the error

Received:

1 0 1 1 1	
0 1 1 1 0 ← parity fails	Therefore:
1 1 1 0 1	Row: 2
0 0 1 1 0	Column: 3
-----	Corrupted bit: (row 2, column 3)
0 0 1 1	Received value: 1
↑	Correct value: 0
column 3 fails	Q6 — Multiple errors

Number of errors	Detection
1	✓ Detected + can correct
2	✓ Detected
3	✓ Detected
4	⚠ Not all 4-error patterns can be detected

That last distinction is specifically stated in your course slides.

C. Checksum

Q7

```

11001100
+ 10101010
-----
1 01110110

```

End-around carry:

01110110	Complement:
+ 1	Sum: 01110111
-----	Checksum: 10001000
01110111	Answer: 10001000

Transmitted:

11001100 10101010 10001000

Q8

```

  11110000
+ 10110101
-----
1 10100101
Add carry:
10100101
      1
-----
10100110

```

Complement:

```

Sum:      10100110
Checksum: 01011001

```

Answer: 01011001

Transmitted:

```

11110000 10110101 01011001

```

Q9

```

10101001
00111001
00011101

```

Adding with one's-complement arithmetic gives:

```

11111111

```

Complement:

```

00000000

```

 **Error-free / accepted**

This matches the checksum receiver example in your slides.

★ **D. CRC Answers**

Q10

Dataword:	1001
Generator:	1011

Generator = 4 bits → append **3 zeros**.

Dividend:	1001000
CRC:	110

Answer

CRC = 110

Codeword = 1001110

Receiver:

$1001110 \div 1011 \rightarrow \text{remainder } 000$

Q11

Dataword:	1101
Generator:	101

Generator = 3 bits → append **2 zeros**.

Dividend:	110100
CRC:	10

Answer

CRC = 10

Codeword = 110110

Q12

Dataword:	101101
Generator:	1101

Append 3 zeros:

Dividend: 101101000

Answer

CRC = 010

Codeword = 101101010

Q13

Dataword: 110101
Generator: 1011

Append 3 zeros:

Dividend: 110101000

Answer

CRC = 111

Codeword = 110101111

Polynomial Questions

Q14

$$x^3 + x + 1$$

Write coefficients for:

x^3	x^2	x^1	x^0
1	0	1	1

Therefore:

Generator = 1011

Degree = 3

CRC = 3 bits

Append = 3 zeros

Q15

$$x^4 + x^2 + 1$$

x^4	x^3	x^2	x^1	x^0
1	0	1	0	1

Therefore:

Generator = 10101

Degree = 4

Append = 4 zeros

CRC remainder = 4 bits

★ **Polynomial trick**

If a power is missing, put a **0**.

Example:

$x^4 + x^2 + 1$ $x^4 \rightarrow 1$ $x^3 \rightarrow 0$ $x^2 \rightarrow 1$ $x^1 \rightarrow 0$ $x^0 \rightarrow 1$	= 10101
--	---------

CRC Receiver

Q16

Received: 1001110
Generator: 1011

Division gives:

Remainder = 000

✅ **No error detected**

Q17

Received: 1101001

Generator: 101

Remainder = 00

 **No error detected**

Q18

Received: 101101001

Generator: 1101

Remainder = 011

 **Error detected**

Combined CRC Problems

Q19 ★★

Dataword: 101011

Generator: 1011

Append:

101011000

Division gives:

CRC = 010

Therefore:

Transmitted codeword = 101011010

Receiver:

101011010 ÷ 1011

Remainder = 000

 **No error detected.**

Q20 ★★

Dataword: 111001

Generator: 1101

Sender

CRC = 000

Codeword = 111001000

Now, for example, flip the **first bit**:

Original: 111001000

Corrupted: 011001000

Receiver divides:

011001000 ÷ 1101

Remainder = 010

✗ Error detected

Your answer can differ if you chose a different bit to flip. What matters is performing CRC on *your* corrupted codeword.

Q21 ★★

Dataword: 1011011

Polynomial:

$x^3 + x^2 + 1$

Polynomial:

$x^3 \quad x^2 \quad x^1 \quad x^0$

1 1 0 1

So:

Generator = 1101

Degree 3 → append 3 zeros:

1011011000

CRC division:

CRC = 001

Therefore:

Codeword = 1011011001

Receiver:

1011011001 ÷ 1101

Remainder = 000

✅ Accepted.

Dataword: 1100101

Q22 ★★

Polynomial:

$x^4 + x + 1$

Generator:

x^4	x^3	x^2	x^1	x^0
1	0	0	1	1
= 10011				

Degree = 4 → append 4 zeros.

Dividend = 11001010000

CRC:

0010

Codeword:

11001010010

Receiver verification:

11001010010 ÷ 10011

Remainder = 0000

✅ Accepted.

Q23 — XOR

1011
XOR
1101

0110

Answer = 0110

Remember:

Same	→ 0
Different	→ 1

Q24

Generator length = **5 bits**

Therefore:

Zeros appended = 4

CRC length = 4 bits

Rule

Generator has n bits → CRC has n-1 bits.

Q25

Dataword: 101101

CRC: 011

Attach the CRC directly to the original dataword:

Codeword = 101101011

Q26

Receiver remainder:

000

☒ **No error detected / accept the frame.**

Q27

Receiver remainder:

010

✗ Error detected / frame is corrupted.**Q28 — Identify the Method**

Situation	Method
A. Add one bit to make number of 1s even	Simple parity
B. Rows + columns have parity	Two-dimensional parity
C. Generator polynomial + modulo-2 division	CRC
D. One's-complement addition + complement	Checksum

☆ CRC Rules to Know Before the Exam

You are given...	Do this
Generator has 4 bits	Append 3 zeros
Generator has 5 bits	Append 4 zeros
x^3+x+1	1011
x^4+x^2+1	10101
CRC arithmetic	XOR
Sender	Divide data + appended zeros
Sender remainder	CRC
Final codeword	Dataword + CRC
Receiver	Divide entire received codeword
Receiver remainder = 000 . . .	☒ No error detected
Receiver remainder $\neq$ 000 . . .	✗ Error detected

The most important trap is: **the sender appends zeros before calculating the CRC, but after finding the remainder, the final codeword is the original dataword followed by the CRC—not the dataword + zeros + CRC.**